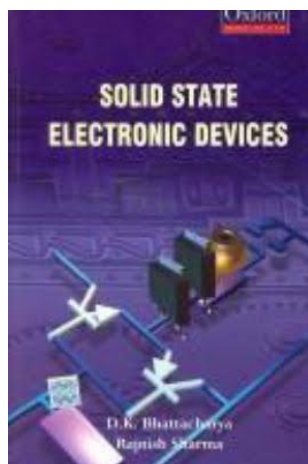


Template for Course Offering

1. Course Title	Physics of Semiconductor devices.
2. Credit Structure	Lecture hours per week: 2 Tutorial hours per week: 1 Practical hours per week: 0 Total Credits: 3 Total Lectures 40 (each lecture @ 1 hr) Total Hours: 40
3. Course Code	EL401
4. Program/Semester	BTech/SEM VI and VII
5. Category	Open Elective
6. Prerequisite courses	Electromagnetism
7. Foundation for	Advanced courses in VLSI Design and Fabrication
8. Abstract Content	The course is designed as per Industry requirements It covers topics related to the working of semiconductor devices, the physics of these devices and device modelling. The course will cover brief on quantum physics, wave-particle duality, silicon as a material for use in VLSI devices. The course has 4 main sections. Section A: Silicon – material and properties. Structure of silicon. Energy Bands Band structure DoS Intrinsic carrier Concentration Doping effects Fermi Dirac Statistics Fermi Level Section B : Transport of electron and holes High field transport Drift and Diffusion Einstein Equation Current Continuity equation PN Junction – under equilibrium, Under bias, Generation and recombination of carriers. Section C: Application of PN junction Schottky Junction BJT transistor Equation and Delays MOS Capacitor MOSFET – working and equations Energy bands of MOSFET Section D : Substrate bias effects, Subthreshold conduction Compound semiconductors Hetero Junctions Intro to nanoscale devices High K dielectrics Metal gate Tx Photodetectors LEDs It This gives a better perspective to IC Design and process engineers.

Detailed Course Contents		
Topic Name	Content (2 -3 lines per 4 – 6 lectures)	No. of lectures
Introduction to Semiconductor Industry, types of semiconductors and usage, markets, products	- Intro to Industry, market size, key companies, products, markets. Global scenario, Indian Scenario -Arena of Semiconductors -	5
Material Properties, Silicon and wafer Growth	Material and properties. Structure of silicon. Energy Bands Band structure DoS Intrinsic carrier Concentration Doping effects Fermi Dirac Statistics Fermi Level -	5
Transport of electron and holes High field transport Drift and Diffusion Einstein Equation Current Continuity	Equilibrium carrier concentration Temperature dependence of carrier Concentration High Doping effects and incomplete ionization Mobility and carrier scattering Low field and high field transport Drift, Diffusion Trap states Current continuity equation	6
PN Junction – under equilibrium, Under bias, Generation and recombination of carriers.	PN Junction PN junction under equilibrium PN Junction under bias 2 PN Junction – generation and re-combination centres Application of PN junction -	6
Schottky diode, Ohmic and Schottky contact Band diagrams	Breakdown of junction CV Profile Schotky Junction Schottky Junction under bias	6

Bipolar devices, Current equations, Small signal model	-Introduction to BJT - Working of BJT Delays Equations - MOS capacitor - MOC- CV	
MOS Capacitor, Three operating regions CV Curve	MOS Capacitor CV Curve Trapped charges	6
MOSFET transistor	MOSFET Operation Current equation MOS digital models Analog Models Scaling Short channel effects	6
Outcomes and Objectives	1. spark an interest in Semiconductors / VLSI 2. To understand physics behind the semiconductor devices 3. To Understand Product development cycle – specifications to qualified product 4. To learn device modelling 5. To learn how to use MOS Models in IC design At the end of the course, students will be able to understand the various Concepts related to operation of semiconductor devices, how to use these devices to develop useful circuits.	
Grading Policy	1 Quiz : 10 marks Mid term examination : 30 marks End term examination: 60 marks Total – 100 Marks Grades will be allotted based on GAUSSIAN DISTRIBUTION and not on absolute total marks. The top 10% students taken from the overall marksheet of 100 marks will be given AA grade, next 10% will be given AB grade. Next 10% will be awarded BB and like that other grades	
VIDEO ABOUT COURSE : The attached video explains about the contents and importance of this course.  Intro to EL401 Physics of Semicondu		

BOOKs :

Notes will be provided

Explanatory Notes

1. Enter a discipline (see master list for guidance) and a first digit followed by xx:
1-4 BTech by year, 5 Mtech, 6 MSc, 7 Mdes
2. Please indicate the program and desired semester in which it is to be offered. In 12, comment on what additional courses in that semester it should not conflict with.
3. Appropriate categories: Core (all programs), technical elective, science elective, open elective (BTech), elective (other programs)
4. Indicate the subsequent courses for which the course would be needed.
5. The Instructor is permitted a certain degree of flexibility in each course. The following ratios of primary to optional content in terms of lecture hours are given as guidelines: Foundation Courses – 75:25, Track Core (Group Elective)/M.Tech. Specialization Core – 66:33, Electives – 50:50. For example, a 3-0-0-3 course is expected to have 40-42 lectures in the semester, and the primary content should specify 30 lecture hours worth of material in a Foundation Level course and only 20 hours of material in an Elective Course. The primary content is what the instructor must cover in the course; the remaining is at the discretion of the Instructor.
6. The outcomes and objectives should give on the one hand the skills and competencies acquired by the student as well as the value of the course towards industry/research/academic objectives.